INCH-POUND

A-A-54878
23 July 1993

COMMERCIAL ITEM DESCRIPTION

PLUG, EAR, HEARING PROTECTION, UNIVERSAL SIZE, VINYL FOAM

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-P-37407B, which is cancelled.

This Commercial Item Description covers requirements for disposable, polyvinyl chloride foam, cylindrical ear plugs suitable for use in hearing protection.

Salient characteristics:

The ear plug shall be fabricated from virgin polyvinyl chloride plastic foam which is non-dermatitic, non-toxic and non-allergenic.

The ear plug shall be smooth, of cylindrical shape, and have a diameter of 0.53 to 0.56 inch and length of 0.71 to 0.85 inch.

The ear plugs shall have a density of 0.080 to 0.108 gm/cc (5.0 to 6.75 lbs/ft ^{3}). The plugs shall have a small cell structure with no cell dimension exceeding 1.27 mm (0.050 inch). The plugs shall be colored yellow.

Performance.

Recovery from compression. When tested as below, the ear plugs shall recover 90 percent of its original diameter in 5 to 60 seconds. One-half of the ear plugs in the inspection lot shall be conditioned at 50 percent ± 5 percent relative humidity and the other half at 100 percent + 0 percent, -5 percent. The temperature shall be 70°F to 75°F (21°C to 24°C) and the time of exposure 24 hours. The ear plug is rolled and compressed between the thumb and forefinger for one minute while compressing it to its smallest diameter within

AMSC N/A FSC 6515

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the first 30 seconds and maintaining that diameter during the last 30 seconds. The ear plug shall then be dropped without delay into a glass tube with an inside diameter that is 90 percent (nominally 0.492 inch) of the ear plug's uncompressed diameter. The tube is held vertically and gently tapped on a hard surface until the ear plug expands to a diameter so that it is held by the tube. The recovery time is the time from the release of the ear plug from the fingers until it is held by the tube.

Sound attenuation. Sound attenuation shall be determined in accordance with the method of ANSI S12.6. The mean attenuation shall exceed the following minimum values:

Frequency (Hz)	Mean Attenuation (dB)
125	26
250	29
500	30
1000	29
2000	33
3150	37
4000	37
6300	38
8000	37

Resistance to heat and humidity. The ear plug when exposed at 70° ± 2°C (158° ± 4°F) and 97 percent ± 3 percent relative humidity for 336 hours ± 2 hours shall not deteriorate as shown by discoloration.

Flamability. The ear plug shall not continue to burn for more than 10 seconds after removal of the flame when tested as below. Furthermore, the ear plug shall not drip flaming particles while burning.

The ear plug shall be tested in accordance with ASTM Standard D568-77 except that the plug shall be suspended by means of a thin wire attached to the plug and the flame shall be applied for five seconds.

Patch test. There shall be no more than one positive reaction in the skin patch test. The skin patch test shall be conducted as follows:

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Nine induction applications of a 25 mm disc of the foam material used in the ear plugs shall be applied with tape on 100 human volunteers during the first three weeks of the test. The nine applications are made on alternate days, Monday, Wednesday and Friday, for 24-hours exposures. The patch areas are examined and scored 24 hours after removal of the patch of each of the nine applications. Following the last applications, the volunteers are "rested" for two weeks before the final 24-hour challenge application. The challenge patch exposure is examined and scored 24 and 72 hours after removal of the patch. A report shall be prepared stating the number of positive and negative reactions. Erythema of a distinct nature is a positive reaction.

Instructions. Complete instructions with illustrations for the use and handling of the ear plugs shall be contained on the container for each pair.

Workmanship. The ear plugs shall be free from defects which detract from its appearance or impair its serviceability. The ear plugs shall be free from objectionable odor.

Unit. Package (PG). One pair of ear plugs shall be packaged in a suitable paperboard container. One package containing 200 individually packaged pairs of ear plugs, as specified, constitutes one unit. The unit package shall be a self-dispensing paperboard box having an adherent backing for attachment to a wall or bulkhead.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for examination purposes shall be one ear plug as specified. Sampling shall be inspection level II with an Acceptable Quality Level (AQL) of 1.0.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for test purposes shall be one ear plug. Sampling shall be inspection level S-2 with an AQL of 1.0.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

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If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number and its respective Item Identification is covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6515-00-137-6345	PLUG, EAR, HEARING PROTECTION, CYLINDRICAL, UNIVERSAL SIZE, YELLOW, 400s

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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project Number: 6515-5513
Location: c:\word\doc gandhi\plug

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54878	2. DOCUMENT DATE (YYMMDD): 930723

3. DOCUMENT TITLE: PLUG, EAR, HEARING PROTECTION, UNIVERSAL SIZE, VINYL FOAM

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Defense Personnel Support Center ATTN: DPSC-MSE	(215)7372870	444-2870
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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